

Natalie sketch

$$\text{distance} = 120 \text{ km}$$

$$\text{cons} = 120 \text{ Wh/km}$$

$$\text{avg } V = 3.7 \text{ V}$$

$$\text{avg } I = 2.5 \text{ Ah}$$

$$\rightarrow 300 \text{ V}$$

$$\rightarrow \text{DoD } 80\%$$

$$\rightarrow \text{cell max} = 4.2$$

$$\text{required energy} = 1 \text{ km} \rightarrow 140$$

$$120 \text{ km} \rightarrow ?$$

$$= 120 \times 140 = 16800$$

$$\text{using } 80\% \text{ of battery } 16800 \div 80\% = 21000 \text{ kWh}$$

$$\text{energy avg} = \text{capacity} \times \text{voltage} = 3.7 \times 2.5 = 9.25$$

$V \rightarrow$ in series

energy per cell

$$\text{required} \rightarrow 300 \text{ V}$$

$$\text{Number of batteries} = \frac{300}{3.7} = 81 \text{ cells in series}$$

cells for nominal V

$$\text{max } V \text{ of batteries} = \frac{300}{4.2} =$$

$$\rightarrow \text{Nom} = 81 \times 3.7 = 299.7 \text{ V}$$

$$\rightarrow \text{max} = 81 \times 4.2 = 340.2 \text{ V}$$

$A \rightarrow$ in parallel

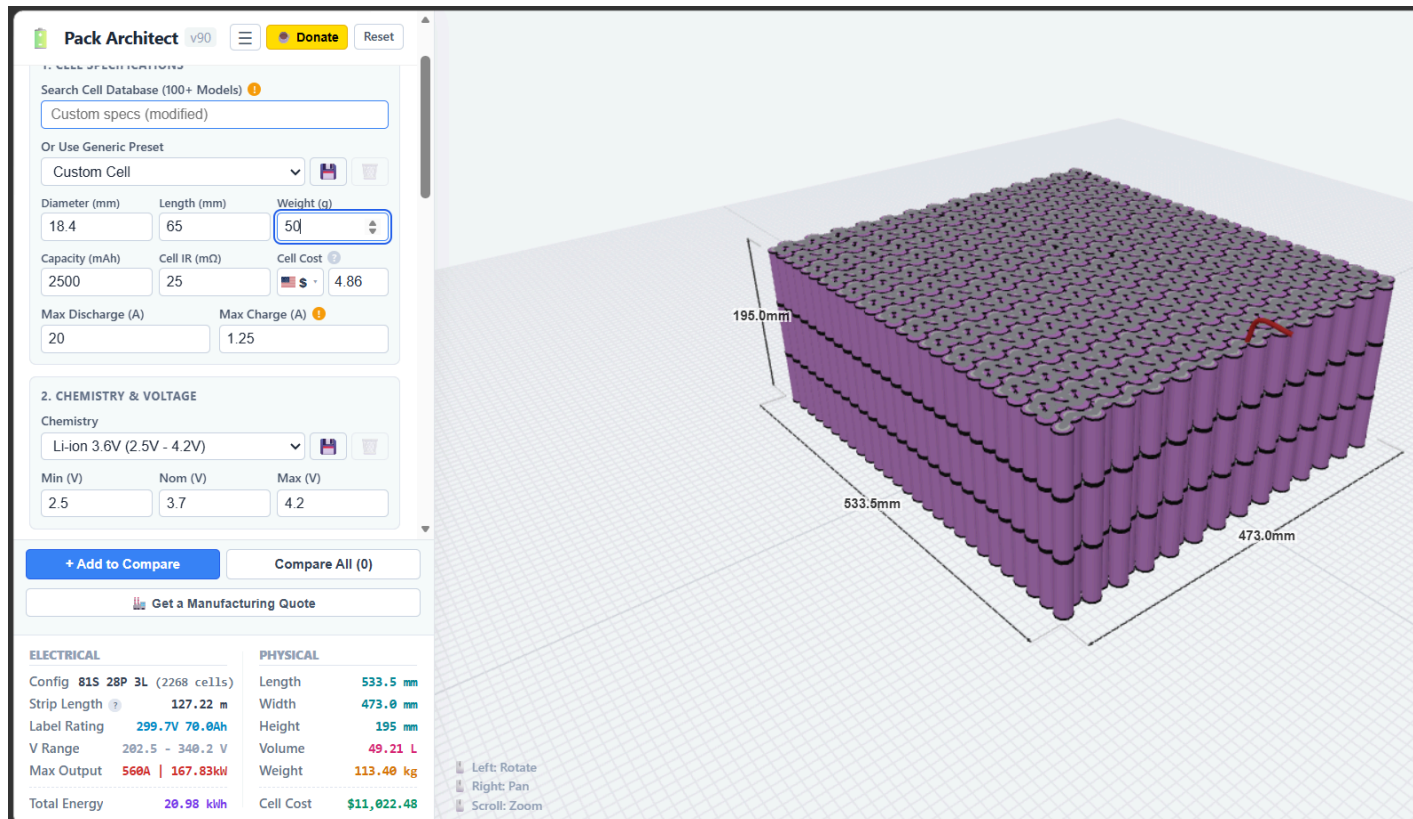
$$\rightarrow \text{Total pack capacity} = \frac{\text{Total pack energy}}{\text{Nominal } V} = \frac{21000}{299.7} = 70.4$$

$$\rightarrow N \text{ in parallel} = \frac{70.4}{2.5} = 28.03 \rightarrow 28 \text{ in parallel}$$

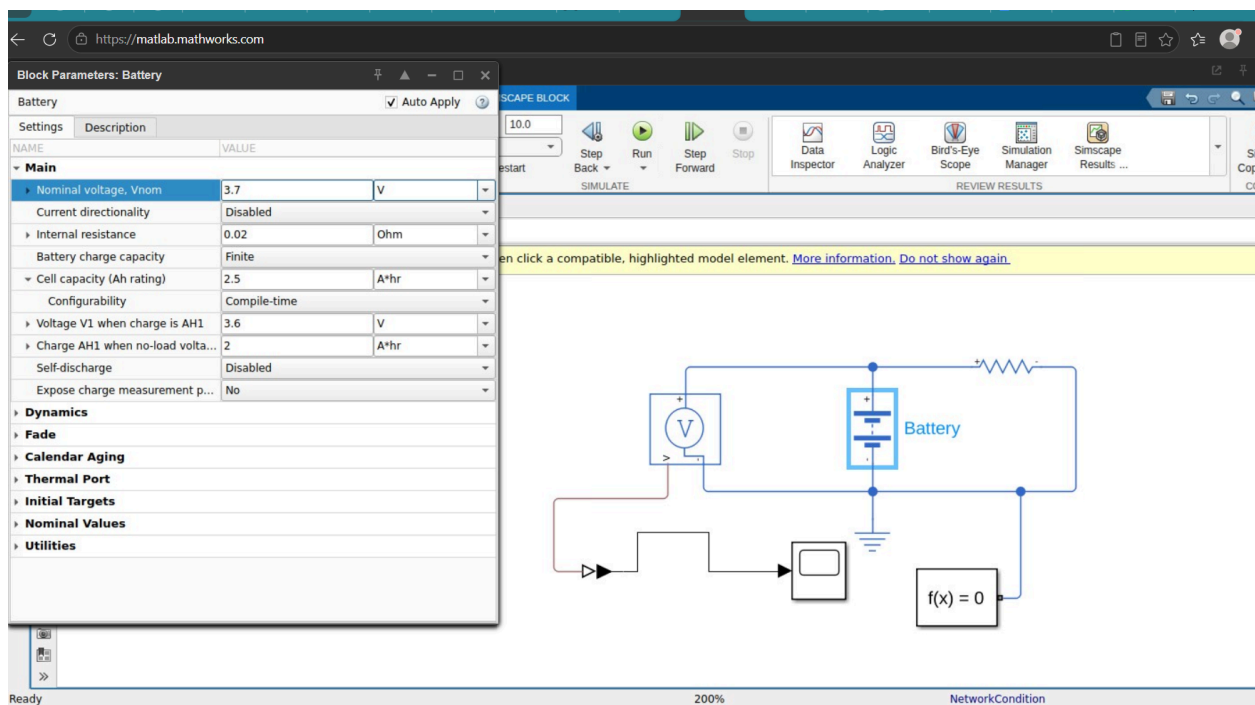
$$\rightarrow \text{Total } N \text{ of cells} = 28 \times 81 = 2268 \text{ cells}$$

$$\rightarrow \text{Total energy} = 2268 \times 9.25 = 20976 \approx 21 \text{ kWh}$$

Good!



Matlab simulation



Scope reading

